

IDA BAGUS DWIWEKA NARATAMA

Seoul, South Korea

Phone : +82 10-8945-7279

[email: dwiwekan@gmail.com](mailto:dwiwekan@gmail.com)

[Linkedin](#) | [Website](#)

ABOUT ME

I'm Nara, an M.S. candidate in Robotics at Kwangwoon University with a strong interest in Artificial Intelligence and how it can connect with robotics and the physical world. My work spans Computer Vision, Multimodal AI, Generative AI, 3D Scene Understanding, SLAM, and Autonomous Navigation, from developing AI models to deploying perception and navigation systems on real robots. I enjoy building intelligent systems that help machines better understand, reason about, and interact with their environment, and commercialize it.

EDUCATION

KWANGWOON UNIVERSITY

M.S in Robotics

GPA: 4.5/4.5

Advisor : Prof. Junghyun Oh

Seoul, South Korea

Feb. 2025 – Expected Feb. 2027

UDAYANA UNIVERSITY

B.S in Information Technology

GPA: 3.93/4.0; Cumlaude

Undergraduate Thesis : Extractive Text Summarization Using a Textual Entailment Approach Based on Deep Learning

Bali, Indonesia

Aug 2019 – April 2023

WORK EXPERIENCE

KWANGWOON UNIVERSITY | ROBOTICS & AI LAB (RAIL)

Graduate Researcher

Seoul, South Korea

Feb 2025 – Present

- Develop AI-based perception, spatial reasoning, and autonomous navigation systems spanning computer vision, multimodal AI, 3D scene understanding, and embodied AI.
- Developed an uncertainty-aware Object Navigation framework using ensemble semantic-map prediction and exploration planning; achieved 64.0% success across 500 HM3D episodes, outperforming deterministic semantic baselines.
- Developed LatentSGC, combining graph neural networks and Conditional Flow Matching for generative 3D Scene Graph Completion; improved Node F1 from 0.245 to 0.432 and Edge F1 from 0.089 to 0.255 against deterministic baselines.
- Reimplemented and deployed a task-driven open-set 3D scene graph pipeline on an AgileX Scout Mini, integrating semantic perception with ROS 2/Nav2 autonomous navigation.
- Built reproducible robot-navigation simulation pipelines using NVIDIA Isaac Sim, ROS 2, SLAM Toolbox, AMCL, and Nav2, with selected configurations achieving 100% navigation success across repeated construction-site trials.

INDONESIA AI (PT. TEKNOLOGI ARTIFISIAL INDONESIA)

Machine Learning Mentor

Depok, West Java, Indonesia

Feb. 2024 – Aug. 2024

- Prepared comprehensive presentations on Time Series analysis, covering fundamental concepts, advanced methodologies, ARIMA, and LSTM models.
- Conducted 12 classes with a total of 20 participants, teaching fundamental machine learning concepts from deep learning to time series.
- Offered personalized one-on-one sessions to explore Machine Learning concepts and applications in depth, tailored to individual learning needs and pace.

PukulEnam.ai

AI Engineer

Bali, Indonesia

Jan. 2024 – Sep. 2024

- Developed a Llama 3-based virtual teaching assistant and explored edge deployment on NVIDIA Jetson Nano.
- Conducted research on Frame Rate Enhancement using DICOM images, developing a model with FiNNiGAN and enhancing it with SRGAN for Super Resolution, achieving superior scores compared to interpolation methods.
- Developed a custom U-Net model to detect anomalies in the ACDC 2017 dataset, optimizing performance through various loss functions, including cross-entropy, focal loss, dice loss, and Lovasz.

- Created and refined code from the OWAD framework to address security issues in datasets, focusing on anomaly detection with benign and malware labels. The goal was to apply OWAD and achieve higher accuracy compared to traditional machine learning models like Random Forest and Decision Trees.
- Developed LLM applications using Llama 3, Retrieval-Augmented Generation (RAG), and the OpenAI API.

BANGKIT ACADEMY

Machine Learning Path Mentor

Indonesia

Feb. 2023 – Jul. 2024

- Mentored and provided educational guidance in Machine Learning, from basic to advanced deep learning, to a total of 75 students across 3 batches (February 2023 to July 2024) from diverse campuses.
- Effectively managed and coordinated in-person Instructor-Led Training (ILT) sessions, collaborating with instructors from various renowned companies to deliver high-quality learning experiences.
- Achieved recognition as a Study Squad Dynamo and Energetic Guide Mentor, being the most active mentor to held study group, sharing session with student about data science, machine learning, and guide to enhance their skill.

NATURAL LANGUAGE PROCESSING RESEARCHER

Udayana University - Profesor's Assistants

Bali, Indonesia

Jul. 2022 – Mar. 2023

- Adapted the Stanford Natural Language Inference (SNLI) dataset into Indonesian and conducted dataset analysis and visualization to support cross-lingual NLP research.
- Built NLP preprocessing and representation pipelines using Stanza, stemming, stopword removal, FastText, GloVe, Word2Vec, and BERT.
- Developed and evaluated deep learning architectures for natural language inference, including BiLSTM, custom interaction layers, Self-Attention, and Multi-Head Attention.

PUBLICATIONS

Ida Bagus Dwiweka Naratama, I Made Putra Arya Winata, Sangcheol Oh, Yuri Song, and Junghyun Oh*, "Transforming Explicit Map Prediction into Uncertainty Exploration for Object Goal Navigation," Manuscript under revision at *International Journal of Control, Automation, and Systems*.

Ida Bagus Dwiweka Naratama, I Made Putra Arya Winata, and Junghyun Oh*, "Latent flow matching for uncertainty-aware 3D scene graph completion from partial robot observations," accepted for presentation at the *International Conference on Control, Automation and Systems*, 2026.

I Made Putra Arya Winata, Donghyun Lee, **Ida Bagus Dwiweka Naratama**, and Junghyun Oh*, "Attention-guided semantic navigation learning from pose, goal text, and visual via exponential reward functions," in *Proceedings of the International Conference on Control, Automation, and Systems*, Incheon, South Korea, Nov. 4–7, 2025, pp. 139–144.

Ida Bagus Dwiweka Naratama, I. M. P. A. Winata, Y. Song, and J. Oh, "VQ-LaMa: Discrete Semantic Map Inpainting for Efficient Exploration," accepted for presentation at *Korea Robotics Society Annual Conference (KRoC)*, February 2026, pp. 858–860.

Ida Bagus Dwiweka Naratama, Song Yuri, Junghyun Oh, "Integrated simulation framework for construction-site quadruped navigation," accepted for presentation at KICEM Annual Conference, Nov. 2025, pp. 273–274.

ADDITIONAL

Language : English (Professional) , Bahasa Indonesia (Native)

Programming : Python, C++, SQL

ML / AI : Pytorch, Transformers, Large Language Models (LLMs), Visual Language Models (VLMs), Generative Modeling, Computer Vision

Robotics : ROS 2, Nav2, SLAM, AMCL, Robot Perception, Autonomous Navigation

Simulation : NVIDIA Isaac Sim, Gazebo, Habitat Simulator

Tools : Git, Linux, Docker, CUDA